TRIG Degrees

[ ]

Rundkorsel (Roundabout) - CAS-first readable

This notebook is structured as short alternating text and math cells.
Goal: explain every step clearly, not only compute answers.

Task (quoted, Danish)

> Projektopgaver: Rundkorsel
>
> Bestem cirklens ligning for vejbanens inderste cirkel,
> hvis koordinatsystemet anbringes i rundkorslens midte.
>
> Bestem cirklens ligning for den yderste cirkel,
> der gar igennem punkterne A (3,38) og B (26,25)
> i det indlagte koordinatsystem.
>
> Hvor lang tid vil det tage at kore en hel tur midt i rundkorslen
> med en anbefalet max hastighed pa 30 km/t?
>
> Bestem koordinaterne til punkterne C og D,
> nar det gives, at Kalundborgvej moder rundkorslen symmetrisk fra B til D.
> Vinklen imellem de to veje er 91 grader.
> Bestem herefter vinklen imellem de to tangenter til cirklen
> i punkterne A og C samt B og D.
>
> Bestem den korteste afstand imellem en bilist midt pa vejbanen
> i rundkorslen og den ventende bla bils midtpunkt P (43,30).
>
> Du skal nu projektere en ny vej,
> hvis linjeforing skal "ramme" rundkorslen fra nordostlig retning.
> Du skal bestemme denne linjeforings rette linje,
> bestemme punktet hvor linjeforingen rammer den yderste cirkel,
> samt de to anlobspunkter ved vejens side,
> der tangerer den yderste cirkel i rundkorslen.
> Du skal antage, at denne nye vej ankommer til rundkorslen
> som Kalundborgvej.

Assumptions used in this notebook:

- `rin = 15`, `rout = 25`, `rmid = 20` (meters).
- `91` is treated as the central angle `AOC`.
- Trig mode is `deg`, so `sin(91)` is in degrees.

[ ]

1) Constants and circle equations in center coordinates

We start with constants and the three concentric circles in local center coordinates `(u,v)`.

[1]

Degrees equation

```
Ax := 3
```

$$Ax = 3$$

```
Ay := 38
```

$$Ay = 38$$

```
Bx := 26
```

$$Bx = 26$$

```
By := 25
```

$$By = 25$$

```
Px := 43
```

$$Px = 43$$

```
Py := 30
```

$$Py = 30$$

```
rin := 15
```

$$rin = 15$$

```
rout := 25
```

$$rout = 25$$

```
rmid := 20
```

$$rmid = 20$$

```
u^2 + v^2 = rin^2
```

$$u^2 + v^2 = 15^2$$

```
u^2 + v^2 = rout^2
```

$$u^2 + v^2 = 25^2$$

```
u^2 + v^2 = rmid^2
```

$$u^2 + v^2 = 20^2$$

[ ]

2) Outer circle in embedded coordinates

From points `A`, `B`, and radius `rout`, there are two candidate centers.
We pick the one with larger `y` (the black dot in the figure is above A and B).

[2]

Degrees equation

```
dx := Bx - Ax
```

$$dx = 26 - 3$$

$$dx = 23$$

```
dy := By - Ay
```

$$dy = 25 - 38$$

$$dy = -13$$

```
Dab := (dx^2 + dy^2)^0.5
```

$$Dab = \left((-13)^2 + 23^2\right)^{0.5}$$

$$Dab = 26.4196896272458$$

```
mx := (Ax + Bx)/2
```

$$mx = \frac{3 + 26}{2}$$

$$mx = \frac{29}{2}$$

```
my := (Ay + By)/2
```

$$my = \frac{25 + 38}{2}$$

$$my = \frac{63}{2}$$

```
a := Dab/2
```

$$a = \frac{26.4196896272458}{2}$$

$$a = 13.2098448136229$$

```
halfChordToCenter := (rout^2 - a^2)^0.5
```

$$halfChordToCenter = \left(-13.2098448136229^2 + 25^2\right.$$

$$halfChordToCenter = 21.2249852767911$$

```
ux := -dy / Dab
```

$$ux = \frac{13}{26.4196896272458}$$

$$ux = 0.492057256667902$$

```
uy := dx / Dab
```

$$uy = \frac{23}{26.4196896272458}$$

$$uy = 0.870562838720134$$

```
h1 := mx + ux*halfChordToCenter
```

$$h_1 = 0.492057256667902 \cdot 21.2249852767911 + \frac{29}{2}$$

$$h_1 = 24.9439080281144$$

```
k1 := my + uy*halfChordToCenter
```

$$k_1 = 0.870562838720134 \cdot 21.2249852767911 + \frac{63}{2}$$

$$k_1 = 49.9776834343563$$

```
h2 := mx - ux*halfChordToCenter
```

$$h_2 = \frac{29}{2} - 0.492057256667902 \cdot 21.2249852767911$$

$$h_2 = 4.05609197188555$$

```
k2 := my - uy*halfChordToCenter
```

$$k_2 = \frac{63}{2} - 0.870562838720134 \cdot 21.2249852767911$$

$$k_2 = 13.0223165656437$$

```
h1N := N(h1, 8)
```

$$h1N = 24.943908$$

```
k1N := N(k1, 8)
```

$$k1N = 49.977683$$

```
h2N := N(h2, 8)
```

$$h2N = 4.056092$$

```
k2N := N(k2, 8)
```

$$k2N = 13.022317$$

```
(h1N, k1N)
```

$$(24.943908,\ 49.977683)$$

```
(h2N, k2N)
```

$$(4.056092,\ 13.022317)$$

```
h0 := h1N
```

$$h_0 = 24.943908$$

```
k0 := k1N
```

$$k_0 = 49.977683$$

```
(x - h0)^2 + (y - k0)^2 = rout^2
```

$$(x - 24.943908)^2 + (y - 49.977683)^2 = 25^2$$

[ ]

3) Find C from angle 91 and D by symmetry

C is built by rotating vector OA by +91 degrees (east-side solution).
D is reflection of B across line OC.

[3]

Degrees expression

```
vAx := Ax - h0
```

$$vAx = 3 - 24.943908$$

$$vAx = -21.943908$$

```
vAy := Ay - k0
```

$$vAy = 38 - 49.977683$$

$$vAy = -11.977683$$

```
vCpx := vAx*cos(91) - vAy*sin(91)
```

$$vCpx = -21.943908\left(-\cos\left(\frac{89\pi}{180}\right)\right) - -11.977683\sin$$

$$vCpx = 21.943908\sin\left(\frac{\pi}{180}\right) + 11.977683\cos\left(\frac{\pi}{180}\right.$$

```
vCpy := vAx*sin(91) + vAy*cos(91)
```

$$vCpy = -21.943908 \sin\left(\frac{89\pi}{180}\right) - 11.977683 \left(-\cos\left(\frac{8}{1}\right.\right.$$

$$vCpy = -21.943908 \cos\left(\frac{\pi}{180}\right) + 11.977683 \sin\left(\frac{\pi}{180}\right.$$

```
vCmx := vAx*cos(91) + vAy*sin(91)
```

$$vCmx = -11.977683 \sin\left(\frac{89\pi}{180}\right) - 21.943908 \left(-\cos\left(\frac{8}{1}\right.\right.$$

$$vCmx = -11.977683 \cos\left(\frac{\pi}{180}\right) + 21.943908 \sin\left(\frac{\pi}{18}\right.$$

```
vCmy := -vAx*sin(91) + vAy*cos(91)
```

$$vCmy = -11.977683 \left(-\cos\left(\frac{89\pi}{180}\right)\right) + 21.943908 \sin\left(\right.$$

$$vCmy = 11.977683 \sin\left(\frac{\pi}{180}\right) + 21.943908 \cos\left(\frac{\pi}{180}\right.$$

```
Cplusx := h0 + vCpx
```

$$Cplusx = \left(21.943908 \sin\left(\frac{\pi}{180}\right) + 11.977683 \cos\left(\frac{\pi}{180}\right)\right) +$$

$$Cplusx = 21.943908 \sin\left(\frac{\pi}{180}\right) + 11.977683 \cos\left(\frac{\pi}{180}\right) + 2$$

```
Cplusy := k0 + vCpy
```

$$Cplusy = \left(-21.943908 \cos\left(\frac{\pi}{180}\right) + 11.977683 \sin\left(\frac{\pi}{180}\right)\right) +$$

$$Cplusy = -21.943908 \cos\left(\frac{\pi}{180}\right) + 11.977683 \sin\left(\frac{\pi}{180}\right) +$$

```
Cminusx := h0 + vCmx
```

$$Cminusx = \left(-11.977683 \cos\left(\frac{\pi}{180}\right) + 21.943908 \sin\left(\frac{\pi}{180}\right)\right)$$

$$Cminusx = -11.977683 \cos\left(\frac{\pi}{180}\right) + 21.943908 \sin\left(\frac{\pi}{180}\right) +$$

```
Cminusy := k0 + vCmy
```

$$Cminusy = \left(11.977683 \sin\left(\frac{\pi}{180}\right) + 21.943908 \cos\left(\frac{\pi}{180}\right)\right) -$$

$$Cminusy = 11.977683 \sin\left(\frac{\pi}{180}\right) + 21.943908 \cos\left(\frac{\pi}{180}\right) +$$

```
N(Cplusx)
```

$$37.3027412033663$$

```
N(Cplusy)
```

$$28.2461569519854$$

```
N(Cminusx)
```

$$13.3510228718687$$

```
N(Cminusy)
```

$$72.1272886965326$$

```
Cx := Cplusx
```

$$Cx = 21.943908 \sin\left(\frac{\pi}{180}\right) + 11.977683 \cos\left(\frac{\pi}{180}\right) + 24.$$

```
Cy := Cplusy
```

$$Cy = -21.943908 \cos\left(\frac{\pi}{180}\right) + 11.977683 \sin\left(\frac{\pi}{180}\right) + 49$$

```
vBx := Bx - h0
```

$$vBx = 26 - 24.943908$$

$$vBx = 1.056092$$

```
vBy := By - k0
```

$$vBy = 25 - 49.977683$$

$$vBy = -24.977683$$

```
dxC := Cx - h0
```

$$dxC = (-1)\,24.943908 + \left(21.943908 \sin\left(\frac{\pi}{180}\right) + 11.977683 \cos\left(\frac{}{18}\right.\right.$$

$$dxC = 21.943908 \sin\left(\frac{\pi}{180}\right) + 11.977683 \cos\left(\frac{\pi}{180}\right.$$

```
dyC := Cy - k0
```

$$dyC = (-1)\,49.977683 + \left(-21.943908 \cos\left(\frac{\pi}{180}\right) + 11.977683 \sin\left(\right.\right.$$

$$dyC = -21.943908 \cos\left(\frac{\pi}{180}\right) + 11.977683 \sin\left(\frac{\pi}{180}\right.$$

```
dotBC := vBx*dxC + vBy*dyC
```

$$dotBC = 1.056092 \left(21.943908 \sin\left(\frac{\pi}{180}\right) + 11.977683 \cos\left(\frac{\pi}{180}\right)\right) - 24.977683 \left(-21.943\right.$$

$$dotBC = -276.0 \sin\left(\frac{\pi}{180}\right) + 560.75752 \cos\left(\frac{\pi}{180}\right.$$

```
dotCC := dxC^2 + dyC^2
```

$$dotCC = \left(21.943908 \sin\left(\frac{\pi}{180}\right) + 11.977683 \cos\left(\frac{\pi}{180}\right)\right)^2 + \left(-21.943908 \cos\left(\right.\right.$$

$$dotCC = 625.0$$

```
projx := (dotBC/dotCC)*dxC
```

$$projx = \frac{\left(-276.0 \sin\left(\frac{\pi}{180}\right) + 560.75752 \cos\left(\frac{\pi}{180}\right)\right)\left(21.943908 \sin\left(\frac{\pi}{180}\right) +\right.}{625.0}$$

$$projx = 7.1994967 \sin\left(\frac{\pi}{90}\right) + 0.52804597 + 10.218476 \text{co}$$

```
projy := (dotBC/dotCC)*dyC
```

$$projy = \frac{\left(-276.0 \sin\left(\frac{\pi}{180}\right) + 560.75752 \cos\left(\frac{\pi}{180}\right)\right)\left(-21.943908 \cos\left(\frac{\pi}{180}\right)\right.}{625.0}$$

$$proj y = -12.488842 - 7.1994967 \cos\left(\frac{\pi}{90}\right) + 10.218476 \mathrm{s}$$

```
vDx := 2*projx - vBx
```

$$vDx = (-1)\, 1.056092 + 2\left(7.1994967 \sin\left(\frac{\pi}{90}\right) + 0.52804597 + 10.2\right.$$

$$vDx = -2.9802322 \cdot 10^{-8} + 14.398993 \sin\left(\frac{\pi}{90}\right) + 20.43695$$

```
vDy := 2*proj y - vBy
```

$$vDy = 2\left(-12.488842 - 7.1994967 \cos\left(\frac{\pi}{90}\right) + 10.218476 \sin\left(\frac{\pi}{90}\right.\right.$$

$$vDy = -14.398993 \cos\left(\frac{\pi}{90}\right) - 2.9802322 \cdot 10^{-8} + 20.43695$$

```
Dx := h0 + vDx
```

$$Dx = \left(-2.9802322 \cdot 10^{-8} + 14.398993 \sin\left(\frac{\pi}{90}\right) + 20.436952 \cos\left(\right.\right.$$

$$Dx = 14.398993 \sin\left(\frac{\pi}{90}\right) + 20.436952 \cos\left(\frac{\pi}{90}\right) + 24.9$$

```
Dy := k0 + vDy
```

$$Dy = \left(-14.398993 \cos\left(\frac{\pi}{90}\right) - 2.9802322 \cdot 10^{-8} + 20.436951 \sin\left(\right.\right.$$

$$Dy = -14.398993 \cos\left(\frac{\pi}{90}\right) + 20.436951 \sin\left(\frac{\pi}{90}\right) + 49.$$

```
N(Cx)
```

$$37.3027412033663$$

```
N(Cy)
```

$$28.2461569519854$$

```
N(Dx)
```

$$45.8709275101925$$

```
N(Dy)
```

$$36.3007007614685$$

```
checkC := simplify((Cx - h0)^2 + (Cy - k0)^2 - rout^2)
```

$$checkC = 0$$

```
checkD := simplify((Dx - h0)^2 + (Dy - k0)^2 - rout^2)
```

$$checkD = -1.2181372 \cdot 10^{-6} \cos\left(\frac{\pi}{90}\right) - 5.8207658 \cdot 10^{-7} - 8.5824681 \cdot 10^{-7} \sin\left(\right.$$

```
checkC
```

$$0$$

```
checkD
```

$$-1.2181372 \cdot 10^{-6} \cos\left(\frac{\pi}{90}\right) - 5.8207658 \cdot 10^{-7} - 8.5824681 \cdot 10^{-7} \sin\left(\frac{\pi}{90}\right) +$$

[ ]

4) Angles between tangents

For points A and C: tangent angle is `180 - 91 = 89`.
For points B and D: compute central angle BOD, then tangent angle is `180 - BOD`.

[4]

Degrees expression

```
phiAC := 180 - 91
```

$$phiAC = 180 - 91$$

$$phiAC = 89$$

```
thetaBD := (180/pi) * acos((vBx*vDx + vBy*vDy)/rout^2)
```

$$thetaBD = \frac{180 \operatorname{acos}\left(\frac{1.056092\left(-2.9802322 \cdot 10^{-8} + 14.398993 \sin\left(\frac{\pi}{90}\right) + 20.436952 \cos\left(\frac{\pi}{90}\right)\right) - 24.977683\left(-14.398\right.}{25^2}\right.}{\pi}$$

$$theta BD = \frac{180 \operatorname{acos}\left(-0.79241767 \sin\left(\frac{\pi}{90}\right) + 1.1406704 \cdot 10^{-9} + 0.6\right.}{\pi}$$

```
phiBD := 180 - thetaBD
```

$$phi BD = 180 - \frac{180 \operatorname{acos}\left(-0.79241767 \sin\left(\frac{\pi}{90}\right) + 1.1406704 \cdot 10^{-9} + \right.}{\pi}$$

$$phi BD = -\frac{180 \operatorname{acos}\left(-0.79241767 \sin\left(\frac{\pi}{90}\right) + 1.1406704 \cdot 10^{-9} + 0.6099\right.}{\pi}$$

```
phiAC
```

$$89$$

```
N(thetaBD)
```

$$54.4120239288058$$

```
N(phiBD)
```

$$125.587976071194$$

[ ]

5) Travel time and shortest distance to P

Lap length on midline is `2*pi*rmid`.
Speed is `30 km/h = 30000/3600 m/s`.
Shortest distance from point `P` to midline circle is `|OP - rmid|`.

[5]

Degrees expression

```
L := 2*pi*rmid
```

$$L = 2\pi 20$$

$$L = 40\pi$$

```
speed := 30000/3600
```

$$speed = \frac{30000}{3600}$$

$$speed = \frac{25}{3}$$

```
T := L / speed
```

$$T = \frac{40\pi}{\frac{25}{3}}$$

$$T = \frac{24\pi}{5}$$

```
OP := ((Px - h0)^2 + (Py - k0)^2)^0.5
```

$$OP = \left((43 - 24.943908)^2 + (30 - 49.977683)^2\right)^{0.5}$$

$$OP = 26.9282433875435$$

```
dmin := abs(OP - rmid)
```

$$dmin = 6.92824338754353$$

```
N(T)
```

$$15.079644737231$$

```
N(dmin)
```

$$6.92824338754353$$

[ ]

6) New road from northeast (45 degrees)

Road centerline goes through `O=(h0,k0)` with slope `tan(45)=1`.
We reuse Kalundborg approach half-angle `alpha = angle(BOC)` to get edge contact points `E1`, `E2`.
Then we compute tangent lines at `E1`, `E2` in slope-intercept form (numeric).

[*]

Degrees *running...*

Click to edit.

[ ]

7) Compact final summary

This final cell prints key numeric results in short lines.
We intentionally skip heavy plotting here to keep notebook layout compact on laptop screens.

[*]

Degrees *running...*

Click to edit.